

# MedEquiSeg: Shared Augmentation and Privileged-Prompt Reliability in Multimodal Medical Image Segmentation

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## Abstract

**Background:** Text can provide target and spatial context for medical image segmentation, but augmentation can invalidate spatial language and target-derived prompts can exaggerate meaningful text use. Accuracy and prompt reliability therefore require separate controls.

**Methods:** MEDEQUISEG combines frozen pooled BioMedCLIP conditioning, two ATConv-equipped projector heads active in the evaluated forward pass,

and deterministic image-mask-text augmentation. The shared policy was introduced retrospectively and entered the complete configuration together with prompt rewriting, which updates direction phrases after flips and right-angle rotations. Five public benchmarks were evaluated with three seeds, native-resolution scoring, five ordered configurations, four prompt controls, and a post hoc presence/class-stratified derangement. A separate three-seed matched BUSI rerun held image-mask augmentation plans fixed and varied only rewriting.

**Results:** MEDEQUISEG achieved a five-dataset macro-average Dice of **87.87%** and IoU of **80.97%**. Versus the horizontal-flip Base reference, Dice was higher by **1.46** points (95% dataset-stratified bootstrap interval **0.95** to **1.99**) and IoU by **1.74** points (95% interval **1.13** to **2.34**). The largest adjacent Dice difference was **+0.84** points between BioMedCLIP plus ATConv and the complete configuration; this transition jointly added shared augmentation and rewriting and did not identify either standalone effect. In the matched BUSI rerun, rewriting minus no rewriting was **+1.17** Dice points (95% hierarchical bootstrap interval **-1.58** to **+4.17**). True text exceeded presence/class-stratified derangement by **+5.63** Dice points (95% interval **+4.71** to **+6.67**), versus **+14.66** under unstratified shuffling.

**Conclusions:** The complete configuration was higher than Base in the ordered analysis, but the sequence does not isolate standalone causal effects. The BUSI rewriting interval included zero, providing no clear accuracy advantage in that dataset and no decomposition of the five-dataset transition. The shuffled-prompt gap was attenuated under one fixed presence/class-stratified assignment, a pattern consistent with coarse class/presence confounding; its residual remains conditional on target-derived prompts and that assignment. These results characterize benchmark performance and prompt sensitivity, not clinical-report understanding or deployment generalization. Code and supporting materials are available at <https://github.com/zjj30/MedEquiSeg>.

**Keywords:** medical image segmentation, multimodal learning, paired augmentation, vision-language models, privileged prompts, prompt reliability

## 1 Background

Vision-language pretraining has made free-text conditioning practical for dense prediction [1–3]. In medical segmentation, language can identify a target, describe morphology, or provide spatial context [4, 6, 7]. This additional modality also creates two reliability requirements that do not exist in image-only segmentation: training transforms must preserve the joint meaning of the image, mask, and prompt, and the resulting prediction should depend on the case-matched prompt when that prompt is claimed to provide useful guidance.

Conventional geometric augmentation applies one transform to the image and mask but leaves text unchanged [14]. If a prompt locates a lesion in the image frame, a horizontal flip changes *left* to *right*, a vertical flip changes *upper* to *lower*, and a right-angle rotation changes both axes. The transformed mask remains geometrically valid, yet the conditioning text may become false. This is an avoidable cross-modal supervision conflict. Separately, a geometrically valid prompt can still be uninformative or weakly grounded; high segmentation accuracy alone cannot show that the text branch is being used. The relevant augmentation example is image-frame location, rather than anatomical laterality: mirroring a breast image does not change the patient’s anatomical left or right.

We therefore treat the retrospective ordered configuration analysis as the main empirical comparison and prompt reliability as a control on how the resulting accuracy is obtained. MEDEQUISEG combines biomedical text features, adaptive convolution, and synchronized image-mask-text augmentation, while within-study ordered configuration contrasts and fixed-model controls constrain interpretation of aggregate performance and language use. Our contributions are:

1. A retrospective comparison of five ordered configurations across five datasets and three random seeds, comprising 75 retained training runs of the base pathway, frozen pooled BioMedCLIP embedding adapter, the effective two-projector ATConv adapter, shared augmentation policy, and prompt rewriting. Because shared augmentation and prompt rewriting enter together only after both adapters, the sequence provides conditional configuration contrasts rather than independent component effects.
2. A comparison under common data partitions and evaluation procedures on five public datasets, with three training seeds, restored native-resolution mask scoring, and three text-guided and six image-only architectures.

3. Evaluation-protocol and prompt-reliability analyses comprising correct/shuffled/fixed/empty inference controls, a post hoc presence/class-stratified prompt derangement, a separately retrained three-seed BUSI matched prompt-rewrite control, prompt-provenance boundaries, and a forward-pass check of which registered ATConv modules enter the forward graph.

## 1.1 Related work

### 1.1.1 Text-guided segmentation

CLIP learns a transferable joint image-text space [1], and CLIPSeg adapts it to dense prediction through a lightweight conditional decoder [2]. ViLT shows that joint vision-language interaction can be built without a region detector or convolutional visual front end [5], while BioMedCLIP introduces biomedical image-text pretraining at scale [3]. LViT integrates language with a transformer-UNet pathway [6]; CausalCLIPSeg uses causal intervention for referring medical segmentation [7]; and MedCLIPSeg uses probabilistic cross-modal adaptation [4]. TextMatch studies multimodal consistency in semi-supervised segmentation [17], whereas ARSeg targets missing or imbalanced textual attributes [18]. These models motivate a controlled test of whether the training prompt still describes the transformed image.

Recent work also separates prompt content from fusion design. DescriptorMedSAM evaluates a structured progression from target name to name-shape-location descriptors and finds that shape and location provide complementary guidance [19]. Spatial-Aware Symmetric Alignment encodes directional constraints as region-level guidance while aligning descriptive and diagnostic expressions [20]. NTP-MRISeg instead casts image, text, and mask tokens as one autoregressive sequence [21]. These architecture- and descriptor-level advances are complementary to MEDEQUISEG, which asks both whether a supplied spatial descriptor remains valid after training-time geometry and whether the trained prediction remains case-specifically text dependent.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.1.2 Image-only reference models</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 185                                                                                                                                                                  |
| U-Net [8], U-Net++ [9], nnU-Net [11], Rolling-Unet [10], U-Mamba [12], and U-KAN [13] demonstrate that a strong visual pathway can perform well without text. A multimodal method should therefore be compared against both text-guided baselines and competitive image-only models. RollingUNet, U-Net, U-Net++, nnU-Net, U-Mamba, and U-KAN are complete for all datasets and all three seeds under the same data partitions and evaluation procedure. This breadth is important because the strongest image-only reference differs across datasets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 186<br>187<br>188<br>189<br>190<br>191<br>192<br>193<br>194<br>195<br>196<br>197<br>198<br>199<br>200<br>201<br>202<br>203<br>204<br>205                             |
| <b>1.1.3 Consistency under paired augmentation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 206                                                                                                                                                                  |
| Data augmentation generally assumes that the target remains valid after a transformation [14]. For image-text learning, this assumption extends to language. Yi et al. update predefined caption keywords when horizontal flipping image-caption pairs and combine this with semantics-preserving appearance augmentation [15]. In multimodal medical segmentation, RBTCA instead grounds textual cues in image regions before augmenting a modality-aware representation [16]. Chai et al. identify the same flip/rotation conflict and avoid it by fusing text into a pseudo-image before augmentation [22]. Semi-MedRef synchronizes positional spans during patch mixing and perturbs position phrases in a semi-supervised teacher-student setting [23]. MEDEQUI-SEG therefore does not claim priority for the general alignment problem, paired caption augmentation, or exchanging individual left/right tokens. Its specific contribution is the cross-dataset evidence chain: an explicit lexical map for a finite image-coordinate vocabulary, a deterministic shared augmentation procedure, an ordered configuration comparison, and a privileged-prompt reliability analysis. | 207<br>208<br>209<br>210<br>211<br>212<br>213<br>214<br>215<br>216<br>217<br>218<br>219<br>220<br>221<br>222<br>223<br>224<br>225<br>226<br>227<br>228<br>229<br>230 |

## 231 2 Methods

232

233 Reporting follows the CLAIM 2024 checklist where applicable [30]; an item-level  
234 checklist accompanies the submission.  
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### 239 2.1 Dataset and prompt provenance

240

241 Table 1 reports the prespecified image-level partitions. The image and mask sources  
242 are BUSI [25], CVC-ClinicDB [26], BUS-BRA [27], BRISC [28], and QaTa-COV19  
243 [29]. Their paired prompts are distributed by the MedCLIPSeg repository [4]. For  
244 lesion cases, MedCLIPSeg extracts class, coarse location, component count, shape, and  
245 tone from the dataset-provided reference mask or class label and fills language-model-  
246 generated templates. Thus these are target-derived synthetic referring captions, not  
247 prospective clinical reports. This benchmark construction is shared by the image+text  
248 methods in Table 4, but it gives them target-derived information unavailable to image-  
249 only methods and limits both clinical interpretation and cross-modality comparison.  
250

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253

### 254 2.2 Reference standards

255

256 The reference standard for each segmentation case is the binary mask distributed  
257 with the corresponding source dataset and retained by the public MedCLIPSeg par-  
258 titution. No reference mask was created, edited, or adjudicated for this study. The  
259 authors verified image-mask pairing, file integrity, split assignment, and the prespec-  
260 ified duplicate-resolution rule, but did not relabel test cases. The original dataset  
261 publications are the source for annotation procedures and annotator qualifications;  
262 details not reported there were unavailable to this secondary analysis. These masks  
263 were selected because they are the released benchmark references used for like-for-like  
264 model evaluation, while possible annotation error and between-annotator variability  
265 remain limitations.  
266

**Table 1** Image-level dataset partitions and realized prompt-change frequencies.

| Dataset  | Train | Val. | Test | Prompt         | Unique | Spatial (%) | Change (%)       |
|----------|-------|------|------|----------------|--------|-------------|------------------|
| BUSI     | 624   | 78   | 78   | Target-derived | 198    | 82.85       | $20.27 \pm 0.10$ |
| ClinicDB | 490   | 61   | 61   | Target-derived | 202    | 100.00      | $28.48 \pm 0.21$ |
| BUS-BRA  | 1311  | 282  | 282  | Target-derived | 75     | 100.00      | $12.44 \pm 0.12$ |
| BRISC    | 3881  | 974  | 892  | Target-derived | 359    | 78.23       | $5.75 \pm 0.04$  |
| COVID-19 | 5716  | 1429 | 2113 | Target-derived | 269    | 100.00      | $45.75 \pm 0.06$ |

*Notes:* Unique and spatial coverage use training prompts. Realized change enumerates all case-epoch events under the fixed 100-epoch shared augmentation policy and reports mean $\pm$ sample standard deviation across seeds 123, 456, and 789. Public prompts are target-derived synthetic captions, not clinical text.

### 2.3 Prompt-evidence scope and real-text testing

The public MedCLIPSeg mask/label-derived captions are restricted to mechanism analysis. They do not constitute real-text testing. Deployment-oriented evidence, which is not available in the present study, would require blinded clinician-authored or clinician-verified prompts generated from images without access to masks. Such a study must freeze the vocabulary and annotation instructions before opening the untouched test set and must compare correct, shuffled, fixed, and empty prompts. We do not use machine-generated captions as a substitute for clinical text evidence.

Spatial coverage counts prompts containing any member of the finite dictionary, including center, middle, top, bottom, upper, lower, left, and right compounds. This count alone overstates intervention frequency because central phrases can be invariant and geometric operations are sampled probabilistically. We therefore apply the deterministic sampling procedure to every training case, epoch, and seed and report the realized proportion of events whose final prompt string differs from the original. Across 100 epochs and seeds 123, 456, and 789, the rates range from  $5.75 \pm 0.04\%$  for BRISC to  $45.75 \pm 0.06\%$  for COVID-19. Complete event counts and operation rates are reported in Supplementary Table S7.

## 323 2.4 Data partitioning and evaluation protocol

324  
325 Dataset partitions were fixed across seeds. Seeds 123, 456, and 789 change parameter  
326 initialization, minibatch order, and augmentation sampling; they do not resplit the  
327 data. The model-optimization partition named Val in the released folders is termed the  
328 validation split here; it selects the best checkpoint. The held-out test split is accessed  
329 only by the final prediction and evaluation scripts. Predictions are restored to the  
330 native reference-mask resolution before scoring.

331  
332 Dataset partitions, split roles, mask-processing rules, random seeds, prediction  
333 threshold, and metric implementation were held constant across the reported com-  
334 parisons. The initial augmentation specification restricted ClinicDB and COVID-19  
335 to horizontal flips and excluded affine transformation, translation, and zoom. The  
336 subsequent shared-augmentation experiment instead applied the same multi-geometry  
337 and appearance policy to all five public datasets, including mild affine transforma-  
338 tion, translation, and zoom. This policy was introduced after the initial augmentation  
339 specification and appears together with prompt rewriting only in the complete con-  
340 figuration. We therefore treat the ordered configuration analysis as a retrospective  
341 follow-up experiment under unchanged data partitions and evaluation procedures,  
342 rather than as a prospectively specified factorial experiment. Implementation records  
343 were retained to support reproducibility (Supplementary Table S11).

344  
345 BRISC contains duplicated images with conflicting masks. Conflicting duplicate  
346 groups were removed according to a prespecified sanitation rule while the original  
347 holdout membership was preserved. The public MedCLIPSeg partition files inherit  
348 the repository’s Train/Val/Test folders and do not expose a populated patient- or  
349 sequence-ID field; consequently, we describe these as fixed image-level holdouts rather  
350 than claiming patient-level separation. A post hoc filename review can recover subject  
351 IDs for part of COVID-19: no recovered subject crosses into the fixed test set, but  
352 434 subjects occur in both its original training and validation folders. We therefore  
353  
354



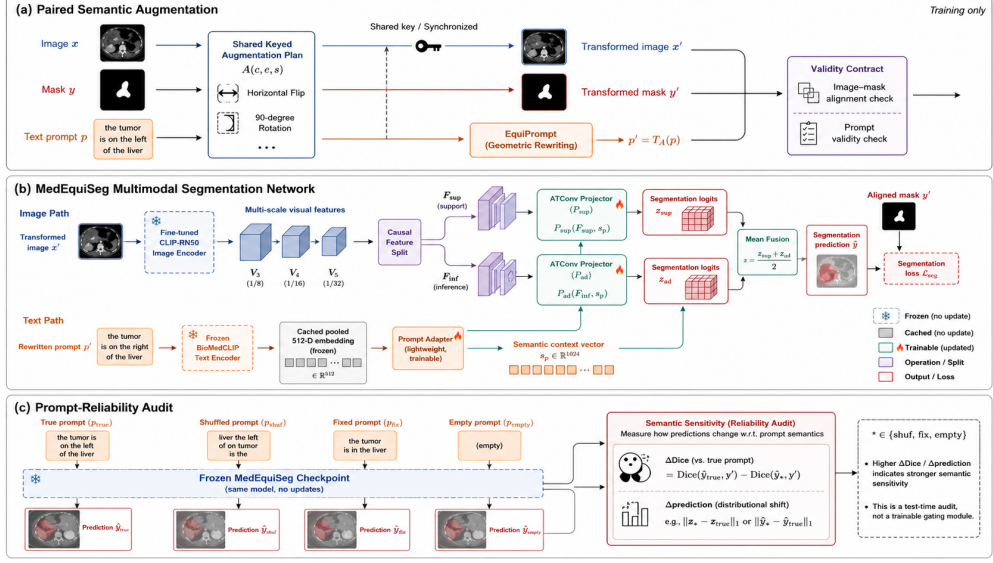
performed a separate sensitivity analysis that assigns every recoverable subject to only one training or validation split while preserving the fixed test set. Rows without recoverable IDs remain case-level groups. The grouped-validation training results are reported in Supplementary Table S10, and the test-set subject-cluster sensitivity based on the reported models additionally addresses uncertainty from repeated test images.

We report Dice and IoU in the main table and additionally export precision, recall, normalized surface Dice (NSD) at two pixels, HD95, and ASSD. A both-empty case receives perfect overlap and zero distance. A one-sided empty case receives zero overlap and infinite distance, and the number of such boundary failures is reported with the distance metrics. Supplementary Tables S1–S5 report NSD, finite-case HD95 and ASSD, and the one-empty failure rate for every method and dataset. Finite distance means are interpreted only together with the failure rate.

## 2.5 MedEquiSeg framework

### 2.5.1 Framework overview and reliability scope

MED EQUISEG comprises a visual segmentation pathway, a frozen biomedical text representation pathway, a shared augmentation policy, and an evaluation-time semantic sensitivity analysis (Fig. 1). The visual pathway fine-tunes the CausalCLIPSeg RN50 encoder and retains its scale-wise supportive/inferior feature partition, and two text-conditioned projection heads. Cached 512-dimensional pooled BioMedCLIP features pass through a trainable state projection and condition both heads; each evaluated head contains one active ATConv operator. During training, the shared augmentation policy keeps the image and mask geometrically aligned and updates discrete spatial phrases when their image-frame meaning changes. During evaluation, correct, shuffled, fixed, and empty prompts test whether the trained model responds to case-specific text. We use *prompt-reliability-aware* in this operational sense: the framework protects



**Fig. 1** Architecture of MEDEQUISEG. A public BUSI training case is used only to illustrate the paired transformation. (A) During training, a deterministic online procedure produces an aligned image-mask pair and the corresponding EquiPrompt lexical view. (B) The fine-tuned CLIP RN50 features are partitioned into supportive and inferior streams, where the latter is the complement of the supportive mask. A trainable state projection maps the cached frozen BioMedCLIP feature to  $s_p$ , which parameterizes the two causal projection heads; each head contains one active ATConv operator. Their logits are averaged before segmentation loss. The transformed target  $y'$  provides the second input to the BCE-Dice loss. Solid modules are optimized, dashed modules are frozen, and the gray feature is cached offline. (C) At evaluation, true, shuffled, fixed, and empty prompts evaluate semantic sensitivity of the same trained model; this analysis is not a learned reliability gate.

and evaluates prompt reliability. It does not estimate per-case uncertainty or contain a learned reliability gate.

### 2.5.2 Shared augmentation policy

The complete configuration uses the shared augmentation policy described above. For case identifier  $c$ , epoch  $e$ , and training seed  $s$ , a deterministic sampling function generates

$$A_{c,e,s} = \mathcal{S}(H, V, R, F, Z, C, G, N \mid c, e, s), \quad (1)$$

where  $H$  and  $V$  are horizontal and vertical flips,  $R$  is a 90-degree rotation,  $F$  is a mild affine transform with translation,  $Z$  is zoom, and  $C$ ,  $G$ , and  $N$  denote color jitter,

gamma, and Gaussian noise. The same realized spatial transformation is applied to image  $x$  and mask  $y$ :

$$x' = A_{c,e,s}^x(x), \quad y' = A_{c,e,s}^y(y). \quad (2)$$

Images use bilinear interpolation, masks use nearest-neighbor interpolation, and the mask is never exposed to appearance operations. The training set is not physically duplicated; a new deterministic transformation is sampled online each epoch.

### 2.5.3 Compositional direction rewriting

Each recognized direction is mapped to a vector  $d = (d_x, d_y)$  with components in  $\{-1, 0, 1\}$ . Horizontal and vertical flips apply

$$T_H(d_x, d_y) = (-d_x, d_y), \quad T_V(d_x, d_y) = (d_x, -d_y), \quad (3)$$

and a counter-clockwise 90-degree rotation applies

$$T_R(d_x, d_y) = (d_y, -d_x). \quad (4)$$

Sequential transforms compose these maps before converting the result back to the original lexical family. Thus, *upper right* remains an atomic target rather than two unrelated replacements. Comma-separated targets are parsed as separate atoms. The BioMedCLIP cache contains the finite lexical closure reachable under all compositions, preventing an augmented prompt from silently falling back to a missing embedding.

### 2.5.4 Augmentation profile and scope

MEDEQUISEG uses one fixed mild augmentation profile:  $p_H = 0.35$ ,  $p_V = 0.10$ , and  $p_R = 0.12$ ; when rotation is selected,  $90^\circ$ ,  $180^\circ$ , and  $270^\circ$  are sampled uniformly. Mild affine is selected with probability 0.15, with angle in  $[-5^\circ, 5^\circ]$  and translation

507 in  $\pm 2.5\%$  of image size. Zoom is selected with probability 0.25 in  $[1.00, 1.07]$ . Color  
508 jitter, gamma, and noise have probabilities 0.45, 0.25, and 0.25, respectively; jitter  
509 magnitude is 0.05, gamma is in  $[0.90, 1.12]$ , and noise standard deviation is 0.006 after  
510 normalization.  
511

512  
513     Only flips and right-angle rotations trigger exact direction rewriting. Small affine  
514 rotations, translation, and off-center zoom do not exchange coordinate axes, but they  
515 can move a target across a coarse location bin. Consequently, MEDEQUISEG guar-  
516 antees lexical consistency for the discrete axis-changing operations, not for arbitrary  
517 continuous spatial descriptions. The small ranges bound this approximation. We addi-  
518 tionally ran MEDEQUISEG + Location Recompute, a benchmark-only diagnostic that  
519 recomputes coarse position phrases from the connected-component centroids of the  
520 transformed training mask whenever affine, translation, or zoom is sampled. Although  
521 it uses the same transform distribution, model family, and training seeds, its augmen-  
522 tation seed key differs from the main run and the realized per-case transforms are  
523 therefore not exactly paired. We report it only as an exploratory same-distribution  
524 diagnostic.  
525  
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### 534 **2.5.5 Multimodal segmentation pathway**

535  
536 The evaluated model initializes the CausalCLIPSeg visual pathway from  
537 the official OpenAI CLIP RN50 checkpoint and fine-tunes it end to end.  
538 Prompts are represented by cached, frozen 512-dimensional pooled features from  
539 `microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224`. A trainable linear  
540 projection maps each pooled feature to the 1024-dimensional state consumed by both  
541 CausalCLIPSeg projectors. Each projector converts this state into a sample-specific  
542  $3 \times 3$  kernel and bias. One same-channel convolution in each active projector is  
543 replaced by ATConv [24], giving two effective ATConv operators in the evaluated  
544 graph (Fig. 2). The adapter also instantiates expanded token features, but all token-  
545 level fusion switches are disabled in the reported configuration; those tokens therefore  
546

do not enter the effective forward path. Implementation metadata record four substitutions because two additional replacements target dormant legacy FPN members that are not called by the released forward implementation; we exclude them from the architectural claim. A forward-pass verification loaded the BUSI model trained with random seed 123 and passed one real test case through the model. Both projector operators were called once with non-zero outputs, while the two registered neck candidates were never called (Supplementary Table S11). Training uses a BCE–Dice objective with Dice weight 0.5. Text conditions visual decoding but does not directly generate the target mask.

## 2.6 Experimental design

### 2.6.1 Training and evaluation

Inputs are resized to  $224 \times 224$  for training. All MEDEQUISEG runs use 100 epochs, batch size 8, AdamW with initial learning rate  $10^{-4}$  and weight decay  $10^{-4}$ , and cosine decay to  $10^{-6}$ . Checkpoint selection uses validation Dice. Evaluation uses a fixed threshold of 0.5. Predictions are restored from the  $224 \times 224$  inference grid to the native reference-mask grid before scoring. The principal MEDEQUISEG and image-baseline environment used Python 3.11.15 and PyTorch 2.5.1 with CUDA 12.1. The nnU-Net/MedCLIPSeg environment used nnU-Net v2.8.0 and MONAI 1.3.0; the official U-Mamba path used Python 3.10.20 and nnU-Net v2.1.1. Runs used NVIDIA RTX 4090 GPUs with driver 580.119.02. A package-level environment record lists the remaining core library versions.

Online complexity is measured from the BUSI models trained with random seed 123 on one NVIDIA GeForce RTX 4090 with batch size 1, FP32, and  $224 \times 224$  inputs. After model-specific warm-up, each forward pass is timed with wall-clock CUDA synchronization; data loading and mask post-processing are excluded. The MEDEQUISEG and LViT measurements include cached-text lookup but not the one-time offline

BioMedCLIP or PubMedBERT encoding step. MedCLIPSeg instead runs its released online UniMedCLIP/BiomedBERT path and is measured with both one probabilistic sample and its official 30-sample test setting. Parameter counts include all modules instantiated in each evaluated model and are not pruned to output-connected sub-graphs. Because the prompt-consistency policy is active only during training, it adds no inference branch or parameter beyond the reported MEDEQUISEG backbone.

In comparative claims, *common data and evaluation procedures* means the same prespecified partitions, split roles, held-out test set, mask sanitation, prediction restoration, and metric implementation. It does not imply that architectures with different official recipes share identical losses, input sizes, or augmentation policies.

### 2.6.2 Baselines and controls

The completed comparison includes MedCLIPSeg, CausalCLIPSeg, LViT, RollingUNet, U-Net, U-Net++, nnU-Net, U-Mamba, and U-KAN [4, 6–13]. The first three and MEDEQUISEG are multimodal; the remaining six models are image-only. Every method has three optimization seeds on every dataset. U-Net, U-Net++, nnU-Net, U-Mamba, and U-KAN contribute 75 dataset–model–seed combinations; RollingUNet contributes 15 more, for 90 image-only training runs in total. The contextual CausalCLIPSeg comparison and the Base reference use the same CLIP-text/horizontal-flip recipe but are independent three-seed training matrices; the latter is the reference for the ordered configuration analysis, whereas the former is retained in the broad model table.

Four inference-time semantic controls are defined without retraining: correct case text, seed-fixed shuffled text, a training-derived fixed prompt, and an empty prompt. Correct text must exceed shuffled text in the Holm-adjusted paired rank screen before a dataset enters the semantic-control visualization; a hierarchical interval excluding zero is additionally required for the evidence to be described as robust to seed uncertainty. The shuffled control applies a deterministic non-zero cyclic shift (random seed 123) to

the ordered test prompts. It preserves the complete prompt multiset and assigns every row another row’s prompt, although duplicated prompt strings can remain textually unchanged; no class or attribute stratum is preserved. To assess this confounding directly, a post hoc fixed-model control permutes the complete prompt multiset separately within target-presence and diagnostic-class strata. Strata are constructed only from released case identifiers and prompt fields; the builder reads neither masks nor image pixels. Within each stratum, a deterministic assignment maximizes changed prompt strings while preserving one-to-one prompt closure. This changes 3,413 of 3,426 public test prompts; 13 BUSI normal cases remain unchanged because every available within-stratum prompt string is identical. Fixed and empty prompts are secondary descriptive controls. The released target-derived captions on the five public datasets define the formal but privileged referring benchmark. Additional prompts generated directly from a case’s mask for oracle diagnostic experiments are a separate category and are excluded from formal performance tables.

The retrospective ordered configuration analysis compares the base Causal-CLIPSeg pathway, BioMedCLIP substitution alone, ATConv alone, their combination, and the complete MEDEQUISEG configuration. Every retained configuration contains all five public datasets and seeds 123, 456, and 789, yielding 75 unique dataset–configuration–seed training runs. The first four configurations use the same horizontal-flip policy. The complete configuration jointly introduces the broader shared augmentation policy and prompt rewriting after both adapters are present; this design does not estimate a standalone effect of either addition or their complete interactions with the adapters. Exploratory runs using other prompt-generation or evaluation definitions are excluded.

We additionally trained a targeted matched prompt-rewrite control on BUSI. For each of seeds 123, 456, and 789, two arms were newly trained for 100 epochs from the same pretrained initialization with the same fixed partition, pooled BioMedCLIP

691 cache, model recipe, and deterministic image-mask augmentation plan. Only the  
692 update of spatial phrases after flips and right-angle rotations was enabled or disabled.  
693 These six runs are separate from the 75-run ordered matrix and estimate neither the  
694 effect of shared augmentation itself nor an across-dataset rewriting effect.  
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697  
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### 700 **2.6.3 Endpoints and statistical analysis**

701  
702 The main accuracy-centered contrast reported in this retrospective ordered configu-  
703 ration analysis is the complete MEDEQUISEG configuration minus the horizontal-flip  
704 Base reference in five-dataset macro-average Dice. This contrast was selected after  
705 model development and is therefore interpreted as post hoc rather than as a prospec-  
706 tively declared primary endpoint. The ordered sequence additionally reports the  
707 intermediate BioMedCLIP and ATConv configurations. Differences among these rows  
708 describe complete configurations within the sequence; they do not estimate inde-  
709 pendent component main effects or complete interactions. The complete-minus-Base  
710 contrast uses case-first paired inference after averaging the three seed values per case.  
711 Dataset-specific Dice and IoU differences and boundary measures are secondary end-  
712 points. The primary semantic-control endpoint is true versus shuffled Dice; fixed and  
713 empty controls are descriptive. The presence/class-stratified derangement is a post hoc  
714 sensitivity analysis of the true-shuffled contrast, not a replacement primary endpoint.  
715 It uses the same case-first three-seed reduction, 10,000 paired bootstrap replicates, and  
716 five-dataset Holm family as the other semantic comparisons. Exactly one precomputed  
717 deterministic within-stratum donor assignment per dataset was evaluated and reused  
718 across the three checkpoints. Its intervals are therefore conditional on those assign-  
719 ments and checkpoints; they do not include randomization uncertainty over other valid  
720 derangements and cannot quantify a causal contribution of the matched attributes.  
721 Cross-model and cross-modality comparisons in Table 4 are contextual rather than  
722 confirmatory because architectures and input information differ.  
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Seed-level results are reported as mean and sample standard deviation. For hypothesis tests, per-case metrics are first averaged across seeds for each method, then compared with paired Wilcoxon signed-rank tests; paired  $t$ -tests are reported as sensitivity analyses. Family-wise multiplicity is controlled with Holm correction [31]. Confidence intervals use 10,000 paired-case bootstrap replicates after the three seed values for each case have been averaged. For the five-dataset macro-average effect, resampling is stratified within each dataset before the five dataset-level effects are averaged, so each benchmark receives equal weight. Training-seed variability is retained in the reported mean and sample standard deviation but is not treated as an additional set of independent cases. Bootstrap intervals estimate the mean effect, whereas the Wilcoxon test targets paired rank differences; the two summaries are interpreted together rather than treated as interchangeable significance decisions. Case-level tests use individual test images as the paired units and are interpreted subject to the patient-grouping limitation described below. Because the COVID-19 test set contains repeated images from 474 recoverable subjects, a post hoc sensitivity analysis retains the image-weighted point estimate but resamples subjects and training seeds for the interval and applies paired tests to equal-subject mean differences. This clustered analysis is reported separately and does not replace the pre-specified image-level analysis. For the targeted BUSI control, rewrite-minus-no-rewrite differences use 10,000 hierarchical bootstrap replicates that resample paired test cases and training seeds. The corresponding case-mean Wilcoxon test is recorded for completeness, but interpretation emphasizes the hierarchical interval because it incorporates the observed variability across the three newly trained seed pairs. As secondary robustness checks using existing predictions only, we also resampled paired test cases and the three training-seed pairs hierarchically and repeated the equal-dataset complete-minus-Base contrast after excluding empty reference masks. These analyses do not retrain a model or turn the three seeds into independent cases (Supplementary Table S16).

### 3 Results

#### 3.1 Segmentation accuracy and ordered configuration contrasts

Across the five public datasets, MEDEQUISEG attains a five-dataset macro-average Dice of 87.87% and IoU of 80.97%. This is 1.34 Dice pp and 1.65 IoU pp above the next reproduced image+text model, CausalCLIPSeg, and 2.91 Dice pp and 2.74 IoU pp above the highest-macro image-only architecture, nnU-Net. These macro comparisons summarize the fixed public benchmark panel; the cross-modality difference remains descriptive because image+text methods receive target-derived captions. Per-dataset results are reported below to expose heterogeneity rather than treating the macro value as a universal ranking.

**Table 2** Five-dataset macro-average accuracy across five retained ordered configurations.

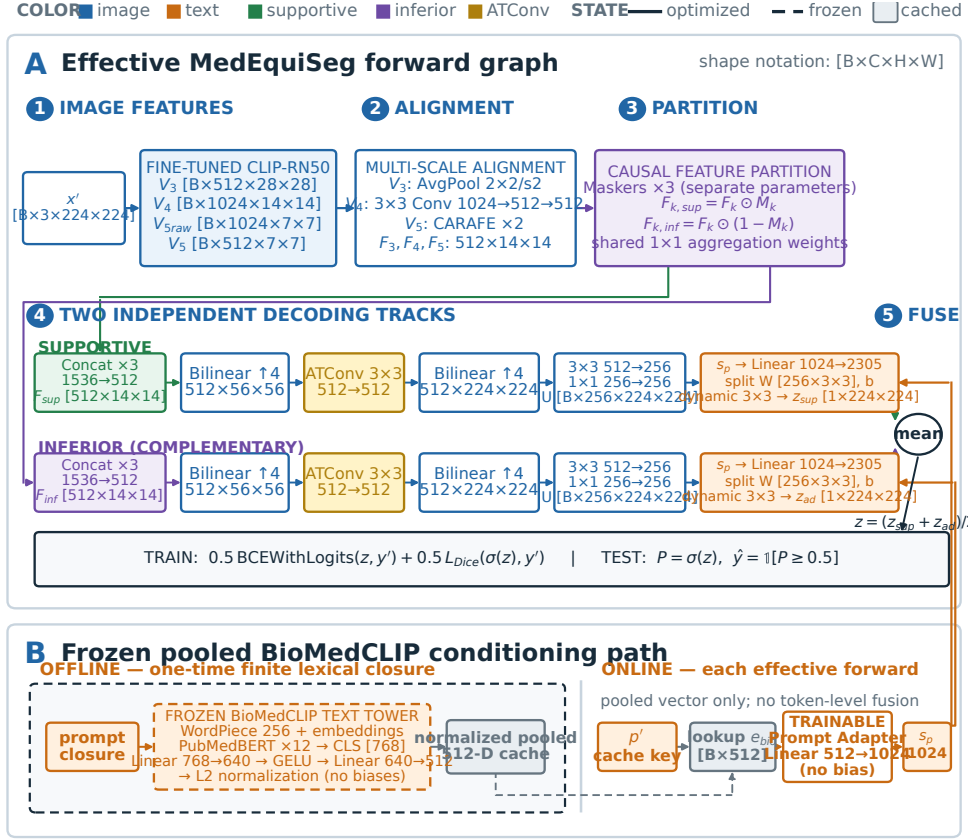
| Configuration            | Text enc. | ATConv | Aug. | Rewrite | Dice             | IoU              |
|--------------------------|-----------|--------|------|---------|------------------|------------------|
| Base model               | –         | –      | –    | –       | $86.41 \pm 0.35$ | $79.22 \pm 0.34$ |
| + BioMedCLIP             | ✓         | –      | –    | –       | $86.67 \pm 0.08$ | $79.52 \pm 0.06$ |
| + ATConv                 | –         | ✓      | –    | –       | $86.63 \pm 0.32$ | $79.42 \pm 0.31$ |
| + BioMedCLIP + ATConv    | ✓         | ✓      | –    | –       | $87.03 \pm 0.15$ | $79.85 \pm 0.23$ |
| <b>MedEquiSeg (Ours)</b> | ✓         | ✓      | ✓    | ✓       | $87.87 \pm 0.17$ | $80.97 \pm 0.13$ |

*Notes:* Five retained configurations, five datasets, and three seeds produce 75 training runs. Values are five-dataset macro-average percentages (mean±sample standard deviation). Text enc. denotes pooled BioMedCLIP conditioning; ATConv denotes the two projection-head operators active in the evaluated forward pass; Aug. denotes the common shared augmentation policy. Shared augmentation and prompt rewriting appear together only in the final configuration, so these are conditional configuration contrasts, not independent component effects or complete interactions. Bold and underline mark the best and second-best configurations. The complete per-dataset matrix is in Supplementary Table S9.

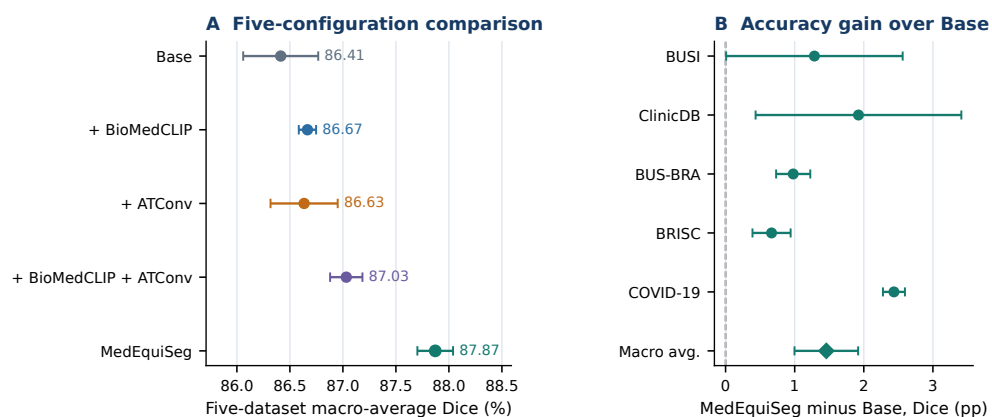
The retrospective comparison includes all five retained configurations across five datasets and three random seeds, for 75 training runs. The five-dataset macro-average Dice is  $86.41 \pm 0.35\%$  for the base pathway,  $86.67 \pm 0.08\%$  for BioMedCLIP alone,  $86.63 \pm 0.32\%$  for ATConv alone,  $87.03 \pm 0.15\%$  for their combination, and  $87.87 \pm 0.17\%$  for the complete MEDEQUISEG configuration. The largest adjacent Dice difference in the displayed sequence is +0.84 pp between BioMedCLIP+ATConv

and the complete configuration. This transition jointly introduces shared augmentation and prompt rewriting and cannot isolate the contribution of either component. The complete-configuration minus Base differences are +1.46 Dice pp (95% dataset-stratified bootstrap CI +0.95 to +1.99) and +1.74 IoU pp (95% CI +1.13 to +2.34). Dataset-level Dice point estimates are positive throughout; only the ClinicDB interval includes zero. These are conditional contrasts between complete configurations. Because shared augmentation and prompt rewriting appear together only in the final row and were not crossed with every BioMedCLIP-ATConv combination, the matrix does not identify either standalone effect or complete component interactions. In the secondary seed-and-case hierarchical bootstrap, the corresponding macro-average intervals were +0.76 to +2.29 Dice pp and +1.02 to +2.55 IoU pp. Excluding the 152 empty reference masks changed the point differences only to +1.53 Dice pp and +1.84 IoU pp (Supplementary Table S16).

In the separate matched BUSI rerun, the shared-augmentation arms without and with prompt rewriting achieved Dice values of  $84.55 \pm 2.03\%$  and  $85.71 \pm 0.57\%$ , respectively. Rewriting minus no rewriting was +1.17 Dice pp (95% hierarchical bootstrap CI -1.58 to +4.17) and +1.45 IoU pp (95% CI -1.55 to +4.79; Supplementary Table S15). Both intervals included zero. This targeted one-dataset control therefore did not show a clear accuracy advantage for rewriting and does not decompose the five-dataset final transition.



**Fig. 2** Effective forward graph of MEDEQUISEG. (A) The fine-tuned CLIP-RN50 supplies three aligned scales to independently parameterized Maskers. The complementary supportive and inferior streams use the same aggregation weights and two independent projectors, each containing exactly one forward-active, visual-only ATConv operator. The two branch logits are averaged before the BCEWithLogits–Soft-Dice training objective or the test-time sigmoid and 0.5 threshold. (B) Offline, the complete frozen BioMedCLIP text tower generates a normalized pooled 512-D cache over the finite prompt closure. Online, the rewritten prompt is a cache key; only the trainable 512-to-1024 adapter follows the lookup. The resulting pooled state  $s_p$  enters only the two dynamic-kernel controllers. Colors identify pathways, whereas solid, dashed, and gray styling denotes optimized, frozen, and cached computation. Layer-resolved modules are shown in Supplementary Figs. S1–S2.



**Fig. 3** Accuracy across five retained ordered configurations under common data partitions and evaluation procedures. (A) Five-dataset macro-average Dice for all five configurations; points are three-seed means and bars are sample standard deviations. (B) Paired three-seed Dice differences between MEDEQUISEG and the Base pathway for each public dataset and their equal-dataset macro; bars are sample standard deviations. The complete-minus-Base contrast is positive across the five-dataset macro. The final configuration transition jointly introduces shared augmentation and prompt rewriting and is not a component-specific effect.

### 3.1.1 Prompt-sensitivity controls

**Table 3** Inference-time prompt controls for trained MEDEQUISEG models on five public datasets.

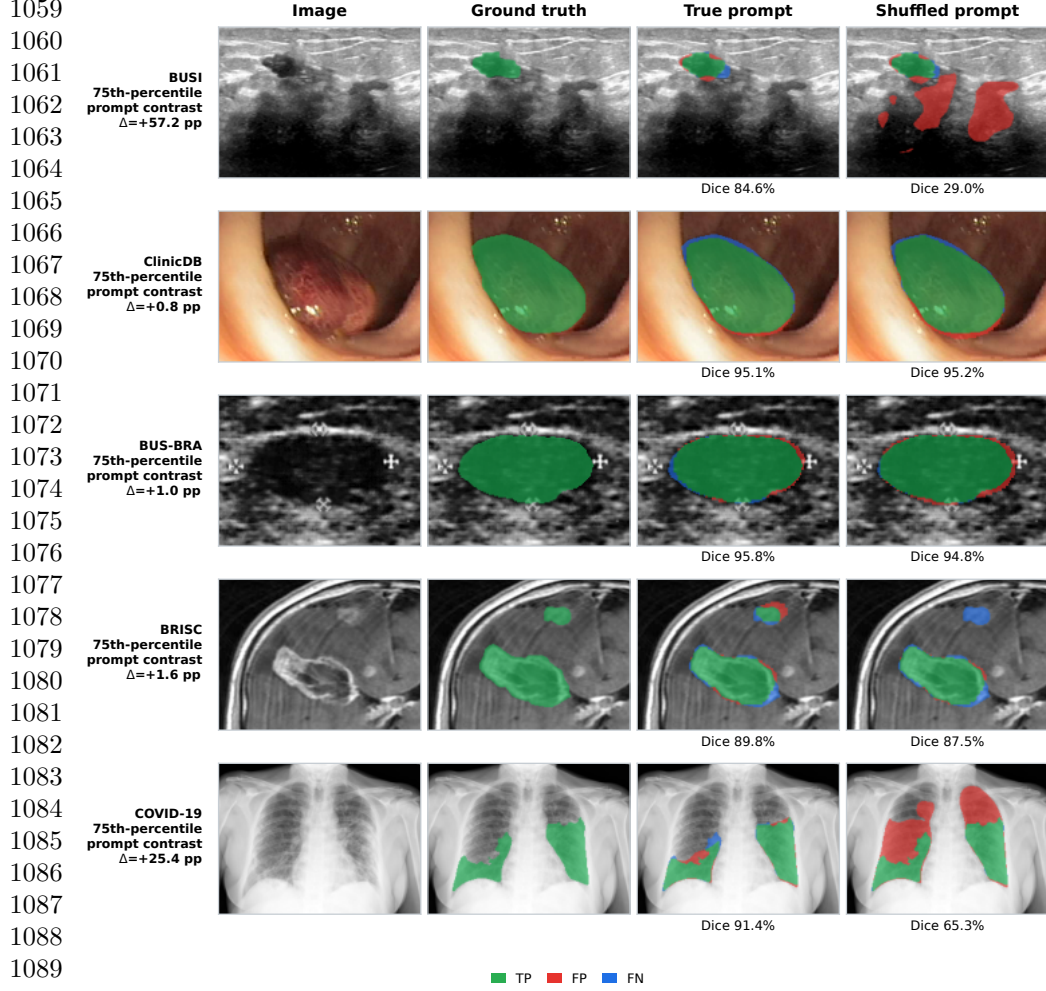
| Dataset  | True Dice        | Shuffled Dice    | Fixed Dice       | Empty Dice       | $\Delta$ pp [95% CI]    |
|----------|------------------|------------------|------------------|------------------|-------------------------|
| BUSI     | $85.78 \pm 0.45$ | $46.73 \pm 1.41$ | $16.69 \pm 0.02$ | $13.22 \pm 5.35$ | $+39.05$ [29.85, 48.49] |
| ClinicDB | $92.07 \pm 0.71$ | $90.20 \pm 0.94$ | $88.31 \pm 1.31$ | $78.71 \pm 7.93$ | $+1.88$ [0.07, 4.22]    |
| BUS-BRA  | $90.24 \pm 0.14$ | $89.47 \pm 0.23$ | $88.87 \pm 0.28$ | $68.22 \pm 8.44$ | $+0.77$ [0.45, 1.15]    |
| BRISC    | $87.18 \pm 0.14$ | $70.89 \pm 2.12$ | $75.31 \pm 1.97$ | $58.89 \pm 5.30$ | $+16.29$ [14.26, 18.43] |
| COVID-19 | $84.08 \pm 0.13$ | $68.80 \pm 0.82$ | $68.54 \pm 0.16$ | $55.81 \pm 3.82$ | $+15.29$ [14.49, 16.10] |

*Notes:* No model is retrained; only the test prompt changes. Dice is mean $\pm$ sample standard deviation over three seeds. The last column is true minus shuffled Dice with a paired-case bootstrap 95% interval after three-seed averaging. Holm-adjusted Wilcoxon tests and the bootstrap intervals are positive on all five datasets.

The semantic controls evaluate whether predictions depend on the supplied prompt. True minus shuffled Dice ranges from +0.77 pp on BUS-BRA and +1.88 pp on ClinicDB to +39.05 pp on BUSI. Holm-adjusted Wilcoxon tests support a true-shuffled difference on all five public datasets, and the paired-case bootstrap intervals are positive throughout; ClinicDB has the smallest interval (+0.07 to +4.22 pp). This demonstrates sensitivity to privileged case-matched caption identity rather than clinical-report understanding or specifically spatial grounding, because the public captions contain target-derived attributes.

Under the single precomputed presence/class-stratified assignment, the gap is attenuated but not eliminated. True text exceeds the stratified derangement by +5.63 Dice pp in the five-dataset macro-average (95% CI +4.71 to +6.67), compared with +14.66 pp under unstratified shuffling. Four of five dataset bootstrap intervals are positive; Holm-adjusted rank support remains for BUSI, BRISC, and COVID-19, whereas BUS-BRA is neutral. Restricting the analysis to the 3,413 cases whose text string actually changes gives +5.87 pp (95% CI +4.87 to +7.03). For the 474 recoverable COVID-19 subjects, equal-subject averaging gives +22.42 pp with a subject-cluster interval of +21.12 to +23.80. This attenuation is consistent with coarse class/presence

|                                                                                           |      |
|-------------------------------------------------------------------------------------------|------|
| confounding in the unstratified control while leaving a heterogeneous residual associa-   | 1013 |
| tion with target-derived prompt identity. It does not identify the causal share explained | 1014 |
| by the matched attributes, and its intervals are conditional on the one fixed assignment  | 1015 |
| per dataset (Supplementary Table S13).                                                    | 1016 |
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**Fig. 4** True-versus-shuffled prompt sensitivity under fixed model parameters. One non-empty case nearest the 75th percentile of the three-seed per-case Dice difference is shown for each dataset that passes the Holm-adjusted rank screen. Predictions are three-seed majority votes. The left-hand selection delta is the mean paired per-case difference across seeds and therefore need not equal the difference between the displayed majority-vote Dice values. Each row uses one shared context crop determined by the union of the reference mask and both predictions. Green denotes true positive, red false positive, and blue false negative. Selection was deterministic; the exact prompts, case identifiers, and uncropped paths are provided in the supplementary selection record.

### 3.2 Per-dataset comparisons and qualitative error analysis

The compact main-text table reports per-dataset Dice and the equal-dataset macro; the complete Dice and IoU matrix is retained in Supplementary Table S14. The model



**Table 4** Per-dataset and equal-dataset macro Dice on five public benchmarks.

| Method                                  | BUSI         | ClinicDB     | BUS-BRA      | BRISC        | COVID-19     | Macro avg.   |
|-----------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <i>Image+text methods</i>               |              |              |              |              |              |              |
| MedCLIPSeg                              | <b>86.13</b> | <u>91.96</u> | 88.47        | 81.36        | <u>82.62</u> | 86.11        |
| CausalCLIPSeg                           | 84.62        | 91.19        | <u>88.87</u> | <u>86.11</u> | 81.85        | <u>86.53</u> |
| LViT                                    | 81.81        | 91.03        | <u>87.66</u> | <u>85.24</u> | 78.50        | <u>84.85</u> |
| <b>MedEquiSeg (Ours)</b>                | <u>85.78</u> | <b>92.07</b> | <b>90.24</b> | <b>87.18</b> | <b>84.08</b> | <b>87.87</b> |
| <i>Image-only contextual references</i> |              |              |              |              |              |              |
| RollingUNet                             | 72.36        | 87.31        | 88.00        | 85.03        | 79.32        | 82.40        |
| U-Net                                   | 61.31        | 85.61        | 87.35        | 85.00        | 78.81        | 79.62        |
| U-Net++                                 | 76.21        | 89.77        | 88.59        | <u>85.94</u> | <b>80.00</b> | 84.10        |
| nnU-Net                                 | 74.99        | <b>93.62</b> | <u>89.80</u> | <b>87.30</b> | 79.07        | <b>84.96</b> |
| U-Mamba                                 | <u>76.63</u> | <u>92.21</u> | <b>90.34</b> | 85.37        | 79.08        | <u>84.73</u> |
| U-KAN                                   | <b>77.10</b> | <u>91.40</u> | 88.56        | 85.45        | <u>79.38</u> | <u>84.38</u> |

*Notes:* Dataset values are three-seed mean Dice percentages; the macro average is the arithmetic mean of the five dataset means. Bold and underline rank methods only within the image+text or image-only panel. Cross-panel differences are descriptive because the public image+text methods receive target-derived synthetic captions unavailable to image-only methods. Complete mean±sample standard deviation Dice and IoU results are in Supplementary Table S14. The five MEDEQUISEG entries use the models from the ordered configuration analysis.

comparison is contextual. MEDEQUISEG has a higher mean Dice than reproduced MedCLIPSeg on ClinicDB (+0.11 pp), BUS-BRA (+1.77 pp), BRISC (+5.82 pp), and COVID-19 (+1.46 pp), and a lower mean on BUSI (−0.35 pp). Relative to the strongest image-only mean selected within each dataset, MEDEQUISEG is higher on BUSI (+8.68 Dice pp) and COVID-19 (+4.08 pp), and lower by 0.10–1.55 pp on the other public datasets. These cross-modality differences are descriptive, not evidence that privileged target-derived text yields a fair deployment advantage.

The COVID-19 grouped train–validation sensitivity preserved the fixed test set and removed all overlap among the 2,130 recoverable subject tokens. In this separate sensitivity checkpoint set, mean Dice was  $83.99 \pm 0.07\%$  for the MEDEQUISEG configuration, compared with  $84.18 \pm 0.13\%$  for its configuration-matched original-split rerun, a change of −0.19 pp. The corresponding U-Net++ change was +0.14 pp. Thus, the known recoverable-subject overlap had little effect on either configuration’s mean Dice. However, rows without recoverable subject tokens cannot be certified as patient grouped (Supplementary Table S10).

The mask-presence analysis (Supplementary Table S8) shows that BUSI contains 65 non-empty and 13 empty test masks. MEDEQUISEG and U-KAN have non-empty Dice values of  $82.94 \pm 0.55\%$  and  $79.70 \pm 0.48\%$ , a difference of +3.24 pp, whereas the overall difference is +8.68 pp. Empty specificity is 100.00% for MEDEQUISEG and  $64.10 \pm 17.76\%$  for U-KAN. Empty-mask cases therefore amplify, but do not fully account for, the BUSI difference. BRISC has the only other mixed test split (753 non-empty and 139 empty masks); the complete subgroup analysis is reported rather than using empty cases as implicit evidence of lesion quality.

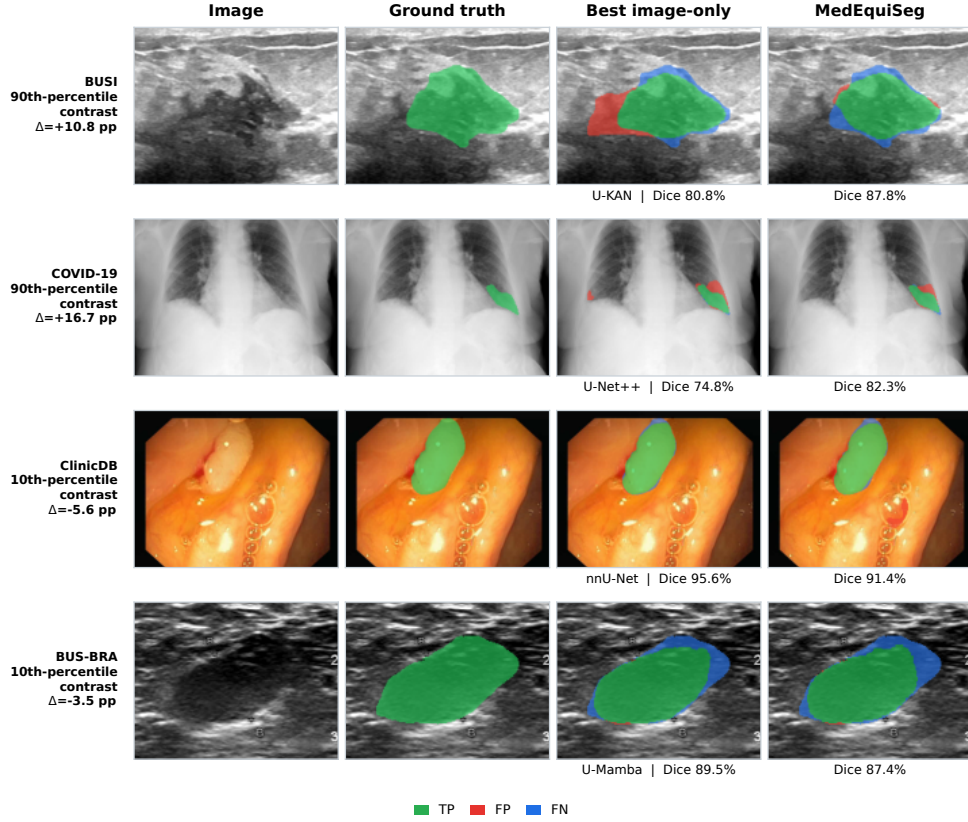
The boundary and empty-mask results in Supplementary Tables S1–S5 follow the same dataset-dependent pattern. Because one-sided empty cases have infinite distance, finite HD95 and ASSD are interpreted only with the failure rate.

**Table 5** Instantiated model size and synchronized single-image inference cost.

| Method                   | Params (M) | Trainable (M) | ms/image     | Images/s | Peak MB |
|--------------------------|------------|---------------|--------------|----------|---------|
| U-Net++ <sup>†</sup>     | 2.41       | 2.41          | 6.80±0.09    | 146.95   | 123     |
| RollingUNet <sup>†</sup> | 1.78       | 1.78          | 26.40±0.65   | 37.88    | 31      |
| LViT                     | 39.93      | 39.93         | 11.35±0.58   | 88.08    | 278     |
| CausalCLIPSeg Baseline   | 165.26     | 165.26        | 18.83±0.53   | 53.10    | 895     |
| MedCLIPSeg ( $S = 1$ )   | 214.58     | 18.68         | 55.44±6.97   | 18.04    | 1814    |
| MedCLIPSeg ( $S = 30$ )  | 214.58     | 18.68         | 816.41±12.91 | 1.22     | 1825    |
| <b>MedEquiSeg (Ours)</b> | 159.78     | 159.78        | 15.03±0.82   | 66.55    | 874     |

*Notes:* Parameter columns count all instantiated modules, not only output-connected subgraphs; trainable counts follow the saved model’s parameter training-state flags. Timing uses batch size 1, FP32,  $224 \times 224$  input, and an NVIDIA GeForce RTX 4090 and is reported as mean±standard deviation after warm-up; data loading and post-processing are excluded. Cached text lookup is included for MEDEQUISEG and LViT, whereas MedCLIPSeg runs its online UniMedCLIP/BiomedBERT towers.  $S$  is the requested number of probabilistic samples. A dagger marks image-only methods.

MEDEQUISEG instantiates 159.78 million parameters and requires  $15.03 \pm 0.82$  ms per image. The training-time policy adds no inference parameters. These measurements describe implementation cost rather than an architecture-controlled speed comparison.



**Fig. 5** Prespecified qualitative error analysis. BUSI and COVID-19 cases are nearest the pre-specified 90th percentile of the three-seed MEDEQUISEG minus image-only Dice difference; ClinicDB and BUS-BRA cases are nearest the 10th percentile. Predictions are pixel-wise majority votes. The left-hand selection delta is the mean paired per-case difference across seeds and therefore need not equal the difference between the displayed majority-vote Dice values. Green is true positive, red false positive, and blue false negative. Within each row, all columns use one context crop determined by the union of the reference mask and both predictions. The deterministic case and prompt record is provided in the supplementary data archive; examples are not additional tests.

The pre-specified upper- and lower-tail examples in Figure 5 make the claim boundary visible: selected BUSI and COVID-19 cases favor MEDEQUISEG, whereas the ClinicDB and BUS-BRA cases expose false-positive or false-negative failure modes.

## 1243 4 Discussion

1244  
1245 The principal empirical result is higher accuracy for the complete configuration than  
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1247 for the base multimodal pathway: +1.46 Dice pp (95% CI +0.95 to +1.99) and +1.74  
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1249 IoU pp (95% CI +1.13 to +2.34) in the ordered configuration analysis. Because the  
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1251 contrast was selected retrospectively and the shared policy was introduced after the  
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1253 initial augmentation specification, it supports an association on this benchmark panel  
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1255 rather than a prospectively confirmed universal effect. MEDEQUISEG also has the  
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1257 highest equal-dataset macro Dice and IoU among the reproduced models, but this  
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1259 does not mean that it is the best method on every dataset. The strongest image-  
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1261 only result is higher on ClinicDB, BUS-BRA, and BRISC, whereas MEDEQUISEG  
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1263 is higher on BUSI and COVID-19. Reporting both macro and per-dataset results is  
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1265 therefore necessary: the macro captures consistency across the benchmark panel, while  
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1267 the dataset table exposes modality-specific failure and competitive margins.

1265 In the ordered configuration analysis, the BioMedCLIP-only and ATConv-only  
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1267 configurations differ from the Base pathway by +0.26 and +0.22 Dice pp, respectively;  
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1269 the configuration containing both differs by +0.62 pp, and the complete configuration  
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1271 differs by +1.46 pp. The largest adjacent displayed difference is +0.84 pp from BioMed-  
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1273 CLIP+ATConv to the complete configuration. This final transition jointly introduces  
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1275 shared augmentation and prompt rewriting. These values are conditional configuration  
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1277 differences, not an additive causal decomposition. In particular, the ordered com-  
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1279 parison does not include the shared policy or rewriting separately with Base or the  
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1281 adapter-only configurations and therefore cannot estimate either standalone contribu-  
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1283 tion or complete interactions. The strongest supported statement is that the complete  
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1285 configuration exceeds Base on the fixed benchmark panel. The forward-pass verifica-  
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1287 tion confirms that ATConv was not globally disabled: two projector operators execute  
1288

and produce non-zero outputs. It also confirms that the other two registered candidates are dormant, so these results support only the effective two-projector adapter and do not show that all four registered ATConv modules were active.

Prompt reliability remains an interpretive control rather than the primary performance claim. Coupled image-mask transformations preserve geometric alignment; the language map addresses prompt validity under a subset of spatial transforms; and semantic controls test whether the trained model depends on case-matched text. The separately retrained BUSI control varies prompt rewriting while holding the current shared image-mask augmentation plans and other recorded settings fixed. Its Dice difference of +1.17 pp has a 95% hierarchical interval from  $-1.58$  to  $+4.17$  pp. The interval includes zero, so this comparison does not show a clear accuracy advantage for rewriting in BUSI; its one-dataset scope also cannot explain the joint transition in the five-dataset matrix. Likewise, segmentation accuracy without shuffled or empty prompts cannot establish reliable text use.

The semantic-control pattern constrains interpretation. Large true-shuffled gaps on all five public sets show that prompt identity affects the prediction. Under the one fixed assignment that matches target presence and diagnostic class, the five-dataset macro-average gap falls from +14.66 to +5.63 Dice pp. This attenuation is consistent with coarse class/presence confounding in the unstratified control but does not estimate its causal contribution or variation across other valid derangements. The remaining association is concentrated in BUSI and COVID-19 and is nearly absent on BUS-BRA. The residual therefore supports sensitivity to privileged case-specific prompt content, not a general claim of language grounding. The result is consistent with the broader paired-augmentation literature rather than a priority claim. Augment the Pairs rewrites predefined caption keywords during horizontal flips, while RBTCA builds a region-grounded modality-aware representation before augmentation [15, 16].

Our added evidence is chiefly diagnostic: the ordered configuration comparison measures the complete system against earlier configurations, while the prompt controls characterize dependence on privileged case-matched text. The matched BUSI control adds a dataset-specific rewriting contrast under a fixed shared policy, but it does not estimate an across-dataset rewriting effect or the effect of shared augmentation itself. The prompt-design evidence from DescriptorMedSAM and the explicit spatial reasoning in SSA reinforce that location language can be useful when it is both informative and correctly grounded [19, 20]. NTP-MRISeg further shows that text can be integrated through a unified generative objective rather than a conditional decoder [21]. Their published values are not inserted into our quantitative tables because their datasets, prompt construction, splits, architectures, and evaluation recipes are not identical to those used in this study. Accordingly, they contextualize the design space but do not provide directly comparable evidence for or against MEDEQUISEG.

The BRISC result requires particular care because duplicate sanitation changes the effective sample count. The sanitation rule is fixed before training and applied to all compared methods.

#### 4.1 Limitations and next evidence

First, the data partitions and evaluation choices were held constant, but the shared augmentation policy was introduced after the initial augmentation specification and therefore represents a retrospective follow-up analysis. The policy changes the original dataset-specific augmentation rules and adds affine transformation, translation, and zoom. The current audited source release cannot be matched bit for bit to the recorded aggregate-run code digest because the corresponding historical source snapshot is unavailable; exact reconstruction of that software state therefore cannot be guaranteed. The summary metadata also omit batch size and worker count, although

both are preserved in supplementary execution records. Second, four ATConv mod-  
 ules are registered in the saved model, but only the two projector modules participate  
 in the implemented forward pass. The reported parameter count includes the two dor-  
 mant neck modules; we therefore do not attribute performance to four active ATConv  
 operators. Third, shared augmentation and prompt rewriting are introduced together  
 in the final ordered configuration. The targeted matched control varies prompt rewrit-  
 ing under the current shared policy, but it is limited to BUSI and three seed pairs, and  
 its hierarchical interval includes zero. It neither estimates an across-dataset rewrit-  
 ing effect nor isolates shared augmentation or interactions with the adapters. Fourth,  
 the current framework evaluates prompt sensitivity at dataset and model level but  
 does not contain a learned per-case confidence gate or automatic image-only fall-  
 back. Fifth, continuous translation and zoom in the reported MEDEQUISEG results  
 do not recompute coarse location labels. The completed three-seed MEDEQUISEG  
 + Location Recompute diagnostic decreases BUSI performance and is neutral on  
 ClinicDB, but it is mask derived and was not paired to the same sampled transforma-  
 tions as the main configuration; it cannot establish the effect of a clinically supplied  
 continuous-language update. Sixth, the public prompts contain attributes extracted  
 from the target masks or labels. This controlled referring benchmark is useful for test-  
 ing equivariance, but it gives image+text methods privileged target metadata and  
 cannot establish clinical utility or a fair deployment advantage over image-only mod-  
 els. The post hoc stratified derangement controls only coarse presence and diagnostic  
 class derived from released identifiers/prompts; it does not match morphology, loca-  
 tion, wording template, or other target-derived attributes and therefore cannot isolate  
 clinical language grounding. Only one deterministic assignment per dataset was evalu-  
 ated, so its intervals are conditional on that assignment and the available checkpoints;  
 uncertainty across alternative valid assignments is unknown. None of the prompts were

1427 authored prospectively by clinicians. Seventh, the five-configuration analysis is com-  
 1428 plete for its ordered sequence but is not a full crossing of the adapters, augmentation  
 1429 policy, and rewriting. It supports conditional configuration contrasts only. The sepa-  
 1430 rately matched BUSI control lies outside this 75-run matrix and does not convert the  
 1431 ordered sequence into a factorial design or support across-dataset causal attribution.  
 1432 Eighth, although six image-only architectures are evaluated with complete three-seed  
 1433 coverage, architectural recipes are not forced to share input size, loss, or augmenta-  
 1434 tion; the comparison is controlled at the data and evaluation levels rather than as  
 1435 an identical-training ablation. The broad model table is therefore contextual, with  
 1436 rankings separated by modality. Ninth, the five public datasets differ in modality, sam-  
 1437 ple count, prompt diversity, and spatial-language prevalence, so an aggregate average  
 1438 would obscure meaningful heterogeneity. Tenth, the released public benchmark meta-  
 1439 data do not generally expose patient or acquisition-sequence identifiers. The fixed test  
 1440 folders are image-disjoint, and the recoverable COVID-19 subject IDs do not cross into  
 1441 test, but patient-level independence cannot be established for every dataset from the  
 1442 released metadata. For COVID-19, the grouped train-validation sensitivity removes  
 1443 all overlap among 2,130 recoverable subject tokens and changes MEDEQUISEG Dice  
 1444 by only  $-0.19$  pp, but 2,951 rows without recoverable tokens remain case-level groups.  
 1445 The subject-cluster sensitivity based on the reported models preserves the direction  
 1446 of the large U-Net++ and shuffled-prompt contrasts, but comparable clustered infer-  
 1447 ence is not possible for the other public datasets from the released metadata. ClinicDB  
 1448 sequence independence also cannot be verified because sequence identifiers are absent.  
 1449 The current manuscript therefore limits its primary claim to a configuration-level  
 1450 benchmark association across the reported ordered sequence, without assigning the  
 1451 gain causally to an individual component. Prospective clinician-authored text and  
 1452 independent cross-domain testing remain future work rather than prerequisites for the  
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reported benchmark comparison. Complete boundary-metric tables, model complexity,  
and qualitative failure panels are provided in the supplement and main manuscript.

## 5 Conclusions

This study provides a retrospective, three-seed comparison of five ordered configurations of shared augmentation and privileged prompts in multimodal medical image segmentation. The complete configuration achieves a five-dataset macro-average Dice of 87.87% and IoU of 80.97% and differs from the horizontal-flip Base reference by +1.46 Dice pp (95% CI +0.95 to +1.99) and +1.74 IoU pp (95% CI +1.13 to +2.34). The largest adjacent displayed difference is +0.84 Dice pp between BioMed-CLIP+ATConv and the complete configuration. Because that transition jointly introduces shared augmentation and prompt rewriting, it does not isolate either standalone effect. In a separate matched BUSI rerun, rewriting minus no rewriting was +1.17 Dice pp (95% CI −1.58 to +4.17); the interval included zero, so this one-dataset control did not show a clear accuracy advantage for rewriting and does not decompose the five-dataset transition. The supported accuracy claim is therefore a configuration-level benchmark association, not a causal component decomposition, and it does not imply that all four registered ATConv modules were active. Case-specific target-derived text affects aggregate predictions after coarse presence/class matching, but the effect is markedly attenuated and does not establish clinical-report understanding. The distinction between the fixed data/evaluation procedures and the later training-policy amendment, the effective two-projector ATConv graph, the joint final transition, and the privileged-prompt control are integral to interpretation of the results.

## 1519 **Declarations**

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## 1522 **Ethics approval and consent to participate**

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1524 Ethics approval and consent to participate were not applicable to this study because

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1526 it was a secondary computational analysis of five publicly available, de-identified

1527

1528 benchmark datasets; no participants were recruited and no new human data were

1529

1529 collected.

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## 1532 **Consent for publication**

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1534 Not applicable. The manuscript contains no identifiable participant information.

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## 1537 **Availability of data and materials**

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1539 The five public datasets are available from their cited sources. The research mate-

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1541 rials available at <https://github.com/zjj30/MedEquiSeg> include the released model

1542

1543 implementation, dataset-partition files with case-level text prompts, model configu-

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1544 rations, three-seed result tables, statistical outputs, figures, and manuscript source.

1545

1546 The archived software version is v1.1.0 ([https://github.com/zjj30/MedEquiSeg/tree/](https://github.com/zjj30/MedEquiSeg/tree/v1.1.0)

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1547 [v1.1.0](https://github.com/zjj30/MedEquiSeg/tree/v1.1.0)); it targets Linux, is implemented in Python, records its package requirements

1548

1549 and computing environment in the repository, and is licensed under Apache-2.0 with-

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1551 out additional restrictions imposed by the authors. The release is the current audited

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1552 implementation, but a bit-for-bit historical source snapshot matching the aggregate-

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1554 run code digest is unavailable, as described in the Limitations and Supplementary

1555

1556 Table S11. Medical images, masks, third-party checkpoints, and raw training logs are

1557

1557 not redistributed.

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1559

## 1560 **Competing interests**

1561

1562 All authors declare that they have no competing interests or relevant financial or

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1564 non-financial relationships to disclose.

|                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                              |
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| <b>Authors' contributions</b>                                                                                                                                                                                                                                                                                                                                                                                          | 1577                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1578                                                                                         |
| XY designed the experimental methodology and collected and curated the datasets.                                                                                                                                                                                                                                                                                                                                       | 1579                                                                                         |
| RX assisted with model training, revised the manuscript, and organized the references.                                                                                                                                                                                                                                                                                                                                 | 1580<br>1581                                                                                 |
| MZ and YZ reviewed the provided annotations and organized the data; they did not create or adjudicate new reference masks. YL contributed to model refinement and manuscript review and editing. YK supervised the overall research direction and critically revised the manuscript. JZ conducted the experiments, led model training, and drafted the manuscript. All authors read and approved the final manuscript. | 1582<br>1583<br>1584<br>1585<br>1586<br>1587<br>1588<br>1589<br>1590<br>1591<br>1592<br>1593 |
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|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1595                                                                                         |
| Not applicable.                                                                                                                                                                                                                                                                                                                                                                                                        | 1596<br>1597<br>1598<br>1599<br>1600                                                         |
| <b>Use of generative artificial intelligence in manuscript preparation</b>                                                                                                                                                                                                                                                                                                                                             | 1601<br>1602<br>1603<br>1604                                                                 |
| OpenAI Codex was used for code-assisted reproducibility checks and English-language editing. All outputs were reviewed by the authors, who take responsibility for the experimental code, numerical results, citations, and final text. No generative model was treated as an author.                                                                                                                                  | 1605<br>1606<br>1607<br>1608<br>1609<br>1610                                                 |

## Abbreviations

ASSD, average symmetric surface distance; ATConv, attentive convolution; BCE, binary cross-entropy; CARAFE, content-aware reassembly of features; CI, confidence interval; CLIP, contrastive language–image pretraining; HD95, 95th percentile Hausdorff distance; IoU, intersection over union; NSD, normalized surface Dice.

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